

RAG Pipeline

Evaluation Report

Model: **gemma4:e2b** · Generated: 02 May 2026, 10:41 PM

Pipeline Configuration

Configuration	
Retrieval mode	hybrid
Reranker	ON
Adaptive weights	OFF
Fusion strategy	RRF
BM25 / Sem weights	0.5 / 0.5
Filtering	OFF
Cross-modal boost	ON
text_k / rerank_k / image_k	10 / 5 / 5
Embed model	BAAI/bge-large-en-v1.5
Image embed model	ViT-L-14 (laion2b_s32b_b82k)
LLM model(s)	gemma4:e2b
Chunk tokens / overlap	384 / 128

Metrics Summary

Retrieval Quality		
M1 Embedding time ↓	13.2963	s
M2 Index size	176 vectors	
M3 Retrieval latency ↓	0.1674	s
■ embed	0.0469	s
■ BM25	0.0034	s
■ fusion	0.0000	s
■ rerank	0.1070	s
M4 Fused Relevance (Text)	0.7439	
M4 Fused Relevance (Image)	0.4283	
M5 Page Coverage@k ↑	89.29	%
M9 Context length	3190	chars
Generation Quality (vs ground truth — 14/14 available)		

M6 ROUGE-1 ↑	0.5005	
M7 ROUGE-2 ↑	0.1847	
M8 ROUGE-L ↑	0.2901	
M10 BLEU ↑	0.1184	
M11 METEOR ↑	0.3295	
M12 BERTScore F1 ↑	0.8752	
M13 FCD ↓	19.04	
M14 Faithfulness ↑	90.54	%
M15 GT Coverage ↑	98.57	%
Performance		
M16 E2E Latency ↓	19.1180	s
M17 Throughput ↑	0.052	q/s
M18 CPU Usage ↓	0.27	%
M19 RAM Usage ↓	0.118	GB
GPU avg	26.43	%
GPU peak	100.00	%

Query 1 of 14

Question

The trajectory diagram of Voyager 1 and Voyager 2 shows clearly diverging paths after Saturn. Using both the trajectory image and the mission text, explain why the two paths diverge at that point, and what specific orbital geometry decision made Voyager 1's post-Saturn path irrecoverable for further planetary encounters.

Retrieved Context

[1] Flight trajectories of Voyager 1 and 2 through the outer solar system (left) and the launch on top of a Titan III-Centaur rocket (right). Images courtesy of NASA.

Taking advantage of this alignment would be two Voyager spacecraft, both beginning their long journeys in 1977. Voyager 2 launched first, on August 20, followed by Voyager 1 on September 5. Both spacecraft would first fly by Jupiter and use that planet's massive gravity to bend their trajectories to then fly by Saturn. Voyager 1 would also be targeted to fly by Saturn's moon Titan, which was known to have a dense atmosphere, a trajectory that would preclude any future planetary flybys. But the option (Source: Voyager Grand Tour PDF.pdf, page 1)

[2] Both Voyagers quietly coasted as they headed toward their next target, the ringed planet Saturn. Once again, Voyager 1 was first, beginning observations in August 1980 before making its closest approach 77,000 miles above Saturn's cloud tops on November 12, and finishing studies in December. Significantly, Voyager 1 made a close flyby of Saturn's large moon Titan, larger than our own Moon and with a dense atmosphere, coming within 4,000 miles of its surface near its South Pole. The Titan encounter changed Voyager 1's trajectory so it could not make any further planetary encounters – thus beginning its long journey out of our solar system. Voyager 2 started its examination of Saturn in June 1981, making its closest approach at 63,000 miles on August 25, and finishing observations in September.

The instruments aboard the Voyagers returned a wealth of new information about Saturn and the cameras returned thousands of images of the planet, its rings and satellites. Among significant discoveries, the Voyagers found that Saturn's rings were far more complex than previously believed, Titan may have lakes of liquid hydrocarbons, Saturn has auroras near its poles much like Earth does, and that there were three previously unknown satellites. (Source: Voyager Grand Tour PDF.pdf, page 3)

[3] Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right). Images are not to scale. Courtesy of NASA.

Two more stops: Uranus and Neptune

Voyager 2's flyby of Saturn set it up to automatically fly by Uranus after a coast of five and a half years. It began to study the Uranian system in November 1985 and flew within 50,600 miles of its cloud tops on January 24, 1986, before concluding its observations in February. In conducting the first ever robotic reconnaissance of Uranus, Voyager 2 observed the planet's atmosphere, satellites and thin ring system, discovering 11 new moons and a previously unknown magnetic field and returning about 8,000 pictures. And then, it was on to Neptune, arriving there after another coast of three and a half years. Beginning its study in June 1989, Voyager 2 flew within just 3,076 miles to Neptune's cloud tops on August 25, concluding its observations in October. That close pass to Neptune allowed detailed observations of its largest moon Triton. During this first ever close up study of Neptune, Voyager 2 returned about 10,000 photographs of the planet, its satellites and dark rings, discovering 6 new moons, a Great Dark Spot on (Source: Voyager Grand Tour PDF.pdf, page 3)

[Image Captions]

[Image 1 Caption] Flight trajectories of Voyager 1 and 2 through the outer solar system (left) and the launch on top of a

Titan III-Centaur rocket (right). Images courtesy of NASA.

Ground Truth Answer

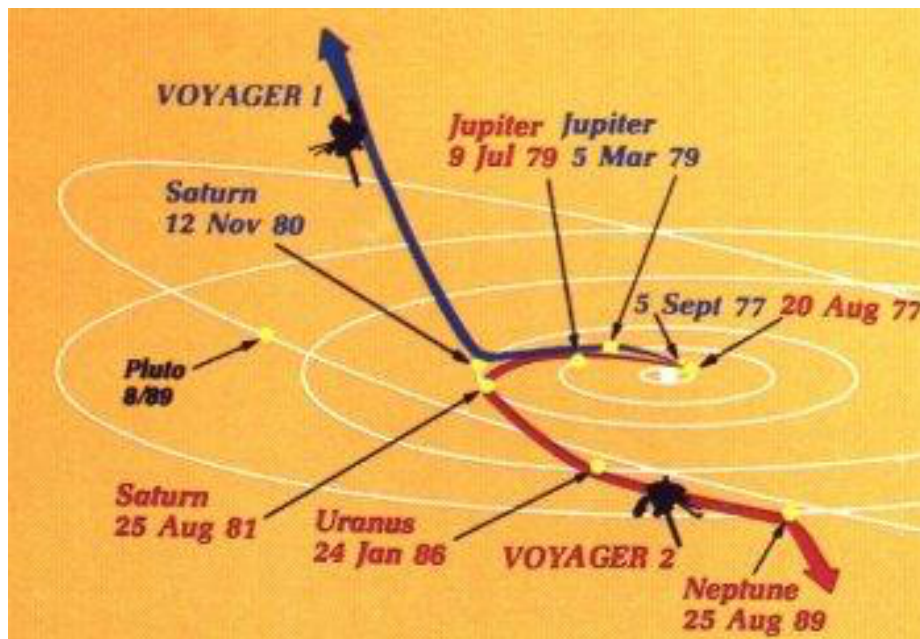
The trajectory diagram shows Voyager 1's path bending sharply away from the ecliptic plane after Saturn while Voyager 2 continues along a shallower arc toward Uranus and Neptune. The text explains this divergence was caused by Voyager 1's close flyby of Titan, coming within 4,000 miles of its surface. That encounter changed Voyager 1's trajectory such that it could not make any further planetary encounters, sending it out of the solar system. Voyager 2 used a different Saturn geometry — a closest approach at 63,000 miles versus Voyager 1's 77,000 miles — that preserved the gravitational slingshot angle needed for the Uranus and Neptune legs. The diagram makes the irreversibility visually concrete: the labeled date nodes on Voyager 2's path continue to Jan 24, 1986 (Uranus) and Aug 25, 1989 (Neptune), while Voyager 1's path terminates its planetary arc at Saturn Nov 12, 1980.

Model Answer

The trajectory diagram shows that both Voyager 1 and Voyager 2 were targeted to fly by Jupiter and then Saturn, using that planet's gravity to bend their trajectories. Voyager 1 was also targeted to fly by Saturn's moon Titan. The text indicates that the encounter with Titan changed Voyager 1's trajectory so that it could not make any further planetary encounters, thus beginning its long journey out of the solar system.

Inference: 27.0356 s · E2E: 27.6404 s · CPU: 0.2% · RAM: 0.381 GB · GPU: 6.3% · Images: 1 · FCD: 15.13

Retrieved Images



Flight trajectories of Voyager 1 and 2 through the outer solar system (left) and the launch on top of a Titan III-Centaur rocket (right).
Images courtesy of NASA.

Query 2 of 14

Question

The Earth-Moon photograph taken by Voyager 1 is described as the first single-frame image of both bodies together. Using the image itself and the surrounding mission text, explain what engineering and operational conditions had to be simultaneously satisfied to produce it, and why this milestone was considered a meaningful preview of capabilities rather than merely a publicity photograph.

Retrieved Context

[1] Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon system in a single frame, giving a sneak preview of the discoveries that lay ahead.

(Source: Voyager Grand Tour PDF.pdf, page 2)

[2] First photograph of the Earth-Moon system in a single frame (left), taken by Voyager 1 on September 18, 1977. Voyager spacecraft as it would appear in space (right). Images courtesy of NASA.

On their way to Jupiter, both Voyagers successfully passed through the Asteroid Belt without incident – two Pioneer spacecraft had made the crossing safely in 1973 and 1974 as pathfinders, proving that it was safe to do so. Voyager 1 arrived first, passing 217,000 miles above the giant planet's cloud tops on March 5, 1979. Two months before the close encounter, Voyager 1 began taking photographs of Jupiter and its moons and gathering other scientific data about the planet and its environment, a process that continued for about a month after the fly-by. Voyager 2 followed not too far behind, beginning scientific observations in April 1979 before passing 350,000 miles above Jupiter's cloud tops on July 9, 1979, and finishing observations about a month later.

The 11 instruments aboard each Voyager made amazing discoveries, not to mention the thousands of stunning photographs of Jupiter and its many moons. Among the more notable discoveries were that Jupiter has a faint ring system, its Great Red Spot is a large swirling storm system, the moon Io has active volcanoes, the moon Europa has a geologically young surface made of ice possibly floating on an ocean of liquid water, and that there were three previously undiscovered small moons.

(Source: Voyager Grand Tour PDF.pdf, page 2)

[3] was kept open, if Voyager 1's Titan flyby was successful, to retarget Voyager 2 to send it on to Uranus and maybe even Neptune – assuming it would survive that long!

Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon system in a single frame, giving a sneak preview of the discoveries that lay ahead.

(Source: Voyager Grand Tour PDF.pdf, page 2)

[Image Captions]

[Image 1 Caption] was kept open, if Voyager 1's Titan flyby was successful, to retarget Voyager 2 to send it on to Uranus

and maybe even Neptune – assuming it would survive that long! Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million

miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon system in a single frame, giving a sneak preview of the discoveries that lay ahead. First photograph of the Earth-Moon system in a single frame (left), taken by Voyager 1 on September 18, 1977. Voyager spacecraft as it would appear in space (right). Images courtesy of NASA.

Ground Truth Answer

The image shows both Earth and the Moon captured together in a single frame from 7.25 million miles, taken just 13 days after launch on September 18, 1977. To produce it, the camera had to achieve stable long-range pointing at two objects separated by significant angular distance, frame both within a single exposure without saturating on

Earth's brightness, and do so while the spacecraft was still in early operational checkout. The text describes it as the first of Voyager 1's many firsts, framing it as a sneak preview of the discoveries ahead. The engineering significance is that it validated the imaging system's ability to handle multi-body framing at interplanetary distances under real flight conditions, not just in laboratory testing. For mission planners, a camera that could jointly frame Earth and Moon from 7.25 million miles had clearly demonstrated the pointing precision and dynamic range that would be needed to image moons near giant planets from comparable or greater distances during the actual science campaign.

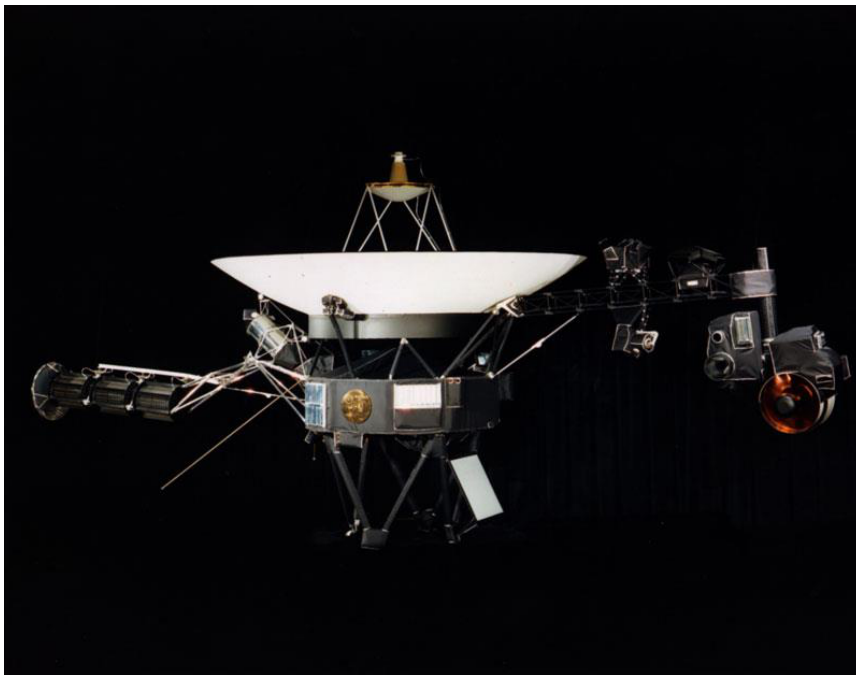
Model Answer

The first photograph of the Earth-Moon system in a single frame was taken by Voyager 1 at a distance of 7.25 million miles, just 13 days after its launch. Producing this image required the successful execution of complex engineering and operational conditions. The text indicates that Voyager 1 turned its camera back toward Earth to capture this photograph. Furthermore, the context mentions that this image gave a "sneak preview of the discoveries that lay ahead," suggesting its significance extended beyond mere publicity.

The milestone was meaningful because it demonstrated the spacecraft's capability to perform a complex maneuver and data acquisition across vast distances. The successful capture of this single frame, combined with the preceding successful navigation and data gathering during the journey, previewed the advanced scientific discoveries that the mission was set to make.

Inference: 13.5002 s · E2E: 13.8445 s · CPU: 0.3% · RAM: 0.060 GB · GPU: 22.8% · Images: 1 · FCD: 16.86

Retrieved Images



was kept open, if Voyager 1's Titan flyby was successful, to retarget Voyager 2 to send it on to Uranus and maybe even Neptune – assuming it would survive that long! Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon system in a single frame, giving a sneak preview of the discoveries that lay ahead. First photograph of the Earth-Moon system in a single frame (left), taken by Voyager 1 on September 18, 1977. Voyager spacecraft as it would appear in space (right). Images courtesy of NASA.

Query 3 of 14

Question

Curiosity's Sky Crane landing image and the text description of the landing sequence together tell a more complete story than either alone. Using both the image of the sky crane lowering Curiosity and the technical description of the EDL sequence, explain why the sky crane approach was necessary rather than airbag-based landing used by earlier rovers, and what physical constraints of the crater site made this the only viable solution.

Retrieved Context

[1] National Aeronautics and Space Administration
Mars Science Laboratory/Curiosity

NASA's Mars Science Laboratory mission set down a large, mobile laboratory — the rover Curiosity — at Gale Crater, using precision landing technology that made one of Mars' most intriguing regions a viable destination for the first time. Within the first eight months of a 23-month primary mission, Curiosity met its major objective of finding evidence of a past environment well suited to supporting microbial life. Curiosity touches down on Mars after being lowered by its Sky Crane.

The rover studies the geology and environment of selected areas in the crater and analyzes samples drilled from rocks or scooped from the surface. The Mars Science Laboratory spacecraft launched from Cape Canaveral Air Force Station, Florida, on Nov. 26, 2011. Mars rover Curiosity carries the most advanced payload of scientific gear ever used on Mars' surface, landed successfully on the floor of Gale Crater on Aug. 6, 2012 Universal Time (evening of Aug. 5, Pacific Time), at 4.6 degrees south latitude, 137.4 degrees east longitude and minus 4,501 meters (2.8 miles) elevation. Its assignment: Investigate whether conditions have been favorable for microbial life and for preserving clues in the rocks about possible past life. Engineers designed the spacecraft to steer it- More than 400 scientists from around the world participate in the science operations. with a series of S-curve maneuvers similar to those used by astronauts piloting NASA space shuttles. During the three minutes before touchdown, the spacecraft slowed its descent with a parachute, then used retrorockets mounted (Source: mars-science-laboratory.pdf, page 1)

[2] It inherited many design elements from them, including six-wheel

In the first few weeks after landing, images from the rover showed that Curiosity touched down right in an area where water once coursed vigorously over the surface. The evidence for stream flow was in rounded pebbles mixed with hardened sand in conglomerate rocks at and near the landing site. Analysis of Mars' atmospheric composition early in the mission provided evidence that the planet has lost much of its original atmosphere by a process favoring loss from the top of the atmosphere rather than interaction with the surface.

In the initial months of the surface mission, the rover team drove Curiosity eastward toward "Yellowknife Bay" to Curiosity's first sample drilling, at a rock called "John Klein."

Mars Science Laboratory/Curiosity

2

NASA Facts

(Source: mars-science-laboratory.pdf, page 2)

[3] capability of landing within a target area only about 20 kilometers (12 miles) long. That precision, about a five-fold improvement on earlier Mars landings, made sites eligible this main destination in September 2014. In the low lay- that would otherwise be excluded for encompassing ers of Mount Sharp during the rover's extended mission, nearby unsuitable terrain. The Gale Crater landing site is so researchers anticipate finding further evidence about hab- close to the crater wall and Mount Sharp that it would not itable past environments and about the evolution of the have been considered safe if the mission were not using Martian environment from a wetter past to a drier present. this improved precision.

Science findings began months before landing. Measure- Curiosity is about twice as long (about 3 meters or 10 feet) ments that Curiosity made of natural radiation levels during and five times as heavy as NASA's twin Mars Exploration the flight from Earth to Mars will help NASA design for as- Rovers, Spirit and Opportunity, launched in 2003. It inher- tronaut safety on future human missions to Mars.

ited many design elements from them, including six-wheel In the first few weeks after landing, images from the rover showed that Curiosity touched down right in an

[Image Captions]

[Image 1 Caption] Mars Science Laboratory/Curiosity Curiosity touches down on Mars after being lowered by its Sky Crane. The Mars Science Laboratory spacecraft launched from Cape Canaveral Air Force Sta-

Ground Truth Answer

The image shows Curiosity being lowered on a tether from the upper sky crane stage to land wheels-down on the Martian surface. The text explains that Curiosity's payload is more than 10 times as massive as earlier rovers and its science equipment required distributing samples to internal analytical instruments, which necessitated landing

upright on wheels rather than rolling in a protective airbag cocoon. The precision landing capability reduced the target ellipse to about 20 kilometers, a five-fold improvement over earlier missions. The Gale Crater site was so close to the crater wall and Mount Sharp that airbag-style landing with the wider uncertainty ellipses of earlier missions would have been unsafe. The sky crane therefore solved two simultaneous problems: it accommodated Curiosity's size and upright instrument layout while also enabling the precision that unlocked Gale Crater as a scientifically viable destination.

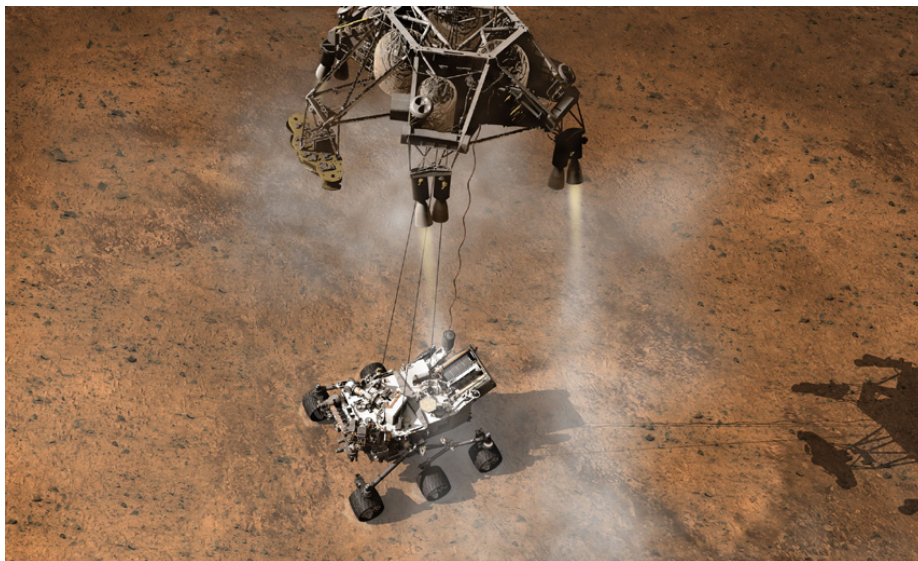
Model Answer

The image of the Sky Crane lowering Curiosity and the accompanying text describe the landing sequence, which together provide a more complete story regarding the mission. The text explains that Curiosity touched down on Mars after being lowered by its Sky Crane, which was used during the descent through Mars' atmosphere. Engineers designed the spacecraft to steer it using a series of S-curve maneuvers, similar to those used by astronauts piloting NASA space shuttles, during this descent.

The text indicates that the spacecraft slowed its descent with a parachute and then used retrorockets mounted on it during the three minutes before touchdown. This process, involving the Sky Crane, was necessary because the mission required precision landing within a target area, which was only about 20 kilometers long. The text implies that this precision capability was achieved by the design elements inherited from earlier Mars rovers, including six-wheel steering, which allowed for a more controlled approach. The context suggests that this method was the only viable solution given the need for this high level of precision for landing within the target area of Gale Crater.

Inference: 18.3782 s · E2E: 18.9035 s · CPU: 0.3% · RAM: 0.072 GB · GPU: 28.7% · Images: 1 · FCD: 29.87

Retrieved Images



Mars Science Laboratory/Curiosity Curiosity touches down on Mars after being lowered by its Sky Crane. The Mars Science Laboratory spacecraft launched from Cape Canaveral Air Force Sta

Query 4 of 14

Question

The image of the rock outcrop called Link and the image of the first drill sample at John Klein represent two distinct evidentiary steps in Curiosity's habitability argument. Using both images and the corresponding text, explain why the visual evidence from Link was scientifically necessary but not sufficient, and what the John Klein drilling contributed that images alone could never provide.

Retrieved Context

[1] investigate an ancient river and fan system identified in The rover analyzed its first scoops of soil on the way to Yellowknife Bay. Once there, it collected the first samples of material ever drilled from rocks on Mars. Analysis of the first drilled sample, from a rock target called "John Klein," provided the evidence of conditions favorable for life in Mars' early history: geological and mineralogical evidence for sustained liquid water, other key elemental ingredients for life, a chemical energy source, and water not too acidic or too salty. On a subsequent drill sample, Curiosity A rock outcrop called Link shows signs of being formed by the deposition of water.

was able to accomplish a first for measurements on another planet: determining the age of the rock. The lowered consideration of more than 30 Martian locations by measurements showed that the drilled material was more than 100 scientists participating in a series of open 4.2 billion years old and yet had been exposed at the workshops. The selection process benefited from examining surface for only 80 million years.

ing candidate sites with NASA's Mars Reconnaissance Orbiter and earlier orbiters, and from the rover mission's In July 2013, Curiosity finished investigations in the Yellowknife Bay area and began a southwestward trek to lowknife Bay area and began a southwestward trek to kilometers (12 miles) long. That precision, about a five-fold the base of Mount Sharp. It reached the base layer of improvement on earlier Mars landings, made sites eligible this main destination in September 2014. In the low lay-that would otherwise be excluded for encompassing ers of Mount Sharp during the rover's extended mission, nearby unsuitable terrain. The Gale Crater landing site is so researchers anticipate finding further evidence about hab- (Source: mars-science-laboratory.pdf, page 2)

[2] ited many design elements from them, including six-wheel In the first few weeks after landing, images from the rover showed that Curiosity touched down right in an area where water once coursed vigorously over the surface. The evidence for stream flow was in rounded pebbles mixed with hardened sand in conglomerate rocks at and near the landing site. Analysis of Mars' atmospheric composition early in the mission provided evidence that the

planet has lost much of its original atmosphere by a process favoring loss from the top of the atmosphere rather than interaction with the surface.

In the initial months of the surface mission, the rover team drove Curiosity eastward toward “Yellowknife Bay” to Curiosity’s first sample drilling, at a rock called “John Klein.” Mars Science Laboratory/Curiosity

2

NASA Facts

(Source: mars-science-laboratory.pdf, page 2)

[3] samples of rocks and soil, distributing them to onboard atmosphere and water. The principal investigator is Paul test chambers inside analytical instruments.

Mahaffy of NASA’s Goddard Space Flight Center, Green-NASA’s Jet Propulsion Laboratory (JPL), Pasadena, Calif., builder of the Mars Science Laboratory, engineered Curiosity to roll over obstacles up to 65 centimeters (25 inches) high and to travel up to about 200 meters (660 feet) arm. It is designed to identify and quantify the minerals in per day on Martian terrain.

rocks and soils, and to measure bulk composition. The principal investigator is David Blake of NASA’s Ames Research Center, Moffett Field, Calif.

ment of Energy radioisotope power generator. The multi-mission radioisotope thermoelectric generator produces Mounted on the arm, the Mars Hand Lens Imager takes electricity from the heat of plutonium-238’s radioactive extreme close-up pictures of rocks and soil, revealing decay. This long-lived power supply gives the mission an details smaller than the width of a human hair. It can also operating lifespan on Mars’ surface of a full Mars year focus on hard-to-reach objects more th

[Image Captions]

[Image 1 Caption] Curiosity is about twice as long (about 3 meters or 10 feet) and five times as heavy as NASA’s twin Mars Exploration Rovers, Spirit and Opportunity, launched in 2003. It inherited many design elements from them, including six-wheel Curiosity’s first sample drilling, at a rock called “John Klein.” Mars Science Laboratory/Curiosity

2

NASA Facts

Ground Truth Answer

The Link outcrop image shows rounded pebbles mixed with hardened sand in conglomerate rock, which the text identifies as evidence for past vigorous water flow — the rounding and sorting of the pebbles indicates sustained fluvial transport. However, visual evidence of ancient stream flow establishes only water presence and movement, not water chemistry, persistence, or geochemical habitability. The John Klein drill image shows the first borehole ever made in Martian rock, and the text explains that analysis of that drilled sample provided the full habitability evidence package: geological and mineralogical evidence for sustained liquid water, key elemental ingredients for

life, a chemical energy source, and water that was neither too acidic nor too salty. Images could establish morphology and sedimentary context; only in situ chemical analysis of drilled interior material could constrain whether the ancient environment was genuinely suited to microbial life.

Model Answer

The visual evidence from the rock outcrop called Link was necessary but not sufficient for the habitability argument because the text states that Link "shows signs of being formed by the deposition of water." However, the text also indicates that the analysis of the first drilled sample from the rock target called "John Klein" provided evidence of conditions favorable for life in Mars' early history, including geological and mineralogical evidence for sustained liquid water, other key elemental ingredients for life, a chemical energy source, and water not too acidic or too salty.

The drilling at John Klein contributed information that images alone could never provide because the drilling allowed for the collection of actual samples of rocks and soils. This physical sampling enabled the analysis that yielded the evidence for liquid water and other key elemental ingredients for life, which is a level of detail beyond what surface visual observation of Link could offer.

Inference: 14.9565 s · E2E: 15.5193 s · CPU: 0.3% · RAM: 0.089 GB · GPU: 28.0% · Images: 1 · FCD: 18.23

Retrieved Images



Curiosity is about twice as long (about 3 meters or 10 feet) and five times as heavy as NASA's twin Mars Exploration Rovers, Spirit and Opportunity, launched in 2003. It inherited many design elements from them, including six-wheel Curiosity's first sample drilling, at a rock called "John Klein." Mars Science Laboratory/Curiosity 2 NASA Facts

Query 5 of 14

Question

The Hubble deep field image and the panel of faint early galaxies shown alongside it are described as evidence for galaxy growth over cosmic time. Using the visual appearance of those galaxies and the text about hierarchical galaxy formation, explain how the morphological evidence in the images connects to the theoretical prediction that large present-day galaxies assembled through mergers rather than forming fully intact.

Retrieved Context

[1] Tracing the Growth of Galaxies

Like documenting a child's development in a scrapbook, astronomers have used Hubble to capture the appearance of many developing galaxies throughout cosmic time. This is possible because of the mathematical relationship between cosmic distance and time: the deeper Hubble peers into space, the farther back it looks in time. As it happens, the most distant and earliest galaxies spied by Hubble are smaller and more irregularly shaped than today's grand spiral and elliptical galaxies. This is evidence that galaxies grew over time through mergers with other galaxies to become the giant systems we see today.

amounts of stars and gas in galaxies, the types and amounts of identifiable chemical elements present, and star-formation rates. And the evolution continues. Hubble observations of our neighboring Andromeda galaxy (M31) have allowed astronomers to predict with certainty that a titanic collision between our Milky Way galaxy and Andromeda will inevitably take place beginning 4 billion years from now. The galaxy is now 2.5 million light-years away, but it is inescapably moving toward the Milky Way under the mutual pull of gravity between the two galaxies and the invisible dark matter that surrounds them both. The merger will likely result in the creation of a giant elliptical galaxy billions of years from now.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 4)

[2] Because the universe was smaller in the past, galaxies were more likely to interact with one another gravitationally. Some of Hubble's cosmic "snapshots" show fantastic stellar streamers pulled out and flung across space by colliding galaxies. They apparently settled over time into the more familiarly shaped galaxies seen closer to Earth and hence nearer to the present time. By carefully studying galaxies at different epochs, astronomers can see how galaxies changed and evolved over time. Among the things they investigate are the relative

Left: Hubble's Ultra Deep Field is one of the most distant looks into space ever. The cumulative exposure time needed to capture the image was about a million seconds (11 days).

Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and frequently interacting in the distant past.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 4)

[3] Viewing Galactic Details and Mergers

When Edwin Hubble discovered that the universe was a vast frontier of innumerable galaxies beyond our Milky Way, he categorized them according to three basic shapes: spiral, elliptical, and irregular. Later, the sharp vision of the space telescope named in his honor revealed unprecedented details in such galaxies. In addition, Hubble's images uncovered a plethora of oddball, peculiarly shaped galaxies that are more numerous the farther back into time the telescope looks. This is because the expanding universe was smaller long ago, and galaxies were both younger and more likely to interact since they were closer to one another.

Among this zoo of odd galaxies are "tadpole-like" objects and apparently merging systems dubbed "train-wrecks." For all their violence, galactic collisions take place at a glacial rate by human standards—timescales on the order of several hundred million years. Hubble's images, therefore, capture snapshots of galaxies at various stages of interaction. The merging of galaxies produces turbulence and tides that can induce new vigorous bursts of star formation within their interstellar gas clouds. These observed mergers form a preview of the coming collision between our own Milky Way and the neighboring Andromeda galaxy 4 billion years from now.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 18)

[Image Captions]

[Image 1 Caption] Left: Hubble's Ultra Deep Field is one of the most distant looks into space ever. The cumulative exposure time needed to capture the image was about a million seconds (11 days). Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and frequently interacting in the distant past.

Ground Truth Answer

The Ultra Deep Field image reveals thousands of galaxies at varying distances, and the close-up panel of faint, distant galaxies shows they are small, irregular, and frequently interacting — very different from the grand spiral and elliptical galaxies seen in the nearby universe. The text states that the most distant and earliest galaxies are smaller and more irregularly shaped, and that the universe was smaller in the past, making gravitational interactions between galaxies more frequent. The visual evidence of disturbed morphologies, asymmetric shapes, and apparent mergers in the distant panels directly supports hierarchical growth: these compact, irregular systems are the raw building blocks that collide and merge over billions of years into the familiar large galaxies seen nearby. The deep field effectively provides a time-ordered fossil record where increasing distance corresponds to increasing lookback time, allowing morphological evolution to be read directly from the images.

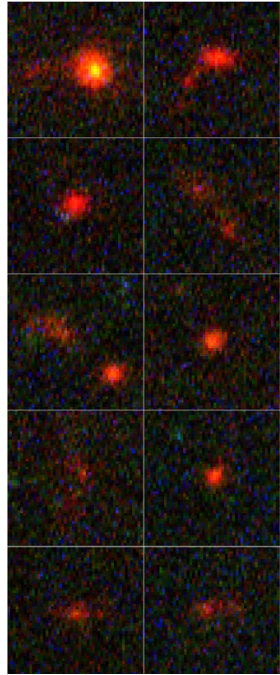
Model Answer

The Hubble deep field image and the sample of faint, early galaxies shown alongside it provide visual evidence supporting the idea that galaxies grew over cosmic time through mergers. The text explains that the most distant and earliest galaxies observed by Hubble are smaller and more irregularly shaped than today's grand spiral and elliptical galaxies. This observation is evidence that galaxies grew over time through mergers with other galaxies to become the giant systems seen today. Furthermore, the text notes that some of Hubble's cosmic "snapshots" show stellar streamers pulled out and flung across space by colliding galaxies, which apparently settled over time into the more familiarly shaped galaxies seen closer to Earth and nearer to the present time. This indicates that the merging of galaxies is a process that shapes the evolution of galaxies. The text also mentions that the merging of galaxies produces turbulence and tides that can induce new bursts of star formation, and these observed mergers form a

preview of future collisions, suggesting that the current large systems assembled through such interactions.

Inference: 15.3974 s · E2E: 15.8282 s · CPU: 0.2% · RAM: 0.090 GB · GPU: 17.4% · Images: 1 · FCD: 18.36

Retrieved Images



Left: Hubble's Ultra Deep Field is one of the most distant looks into space ever. The cumulative exposure time needed to capture the image was about a million seconds (11 days). Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and frequently interacting in the distant past.

Query 6 of 14

Question

The dark matter page shows two views of galaxy cluster CI 0024+17 — one in visible light showing blue arcs, and one with a blue dark matter density overlay. Using both images together with the gravitational lensing text, explain the full chain of inference that connects the visible arc morphology in the first image to the quantitative dark matter mass estimate cited in the text.

Retrieved Context

[1] Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112.

Bottom: These two pictures show a massive galaxy cluster, CI 0024+17. The visible-light image (left) shows blue arcs among the yellowish

galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue

overlay (right) shows the dark matter density needed to account for the gravitational distortions.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[2] Shown here are two Hubble views of the galaxy M84. Left: This camera view shows the bright core at the center of the galaxy surrounded by

a (vertical) dark band of gas and dust. Right: This plot was generated by passing light near the core of the galaxy through a spectrograph. The

thin, vertical rectangle in the center of the left panel shows the size and shape of the spectrograph's sampling slit.

Spectra taken left and right

of the slit's position, which is centered on the core of the galaxy, show a dramatic shift in color to blue or red.

Blueshifted light indicates that the

emitting light source is moving toward Earth, while redshifted light indicates objects that are moving away. At

880,000 miles per hour, stars and

glowing gases nearest to the core of M84 are moving the fastest. They are circling a black hole at the center of the galaxy, and so appear to be

moving rapidly toward Earth on the left and receding on the right.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 7)

[3] Shining a Light on Dark Matter

Dark matter is an invisible form of matter that makes up most of the universe's mass and creates its underlying structure. Dark

matter's gravity drives normal matter (gas and dust) to collect and build up into stars and galaxies. Although astronomers cannot

see dark matter, they can detect its influence by observing how the gravity of massive galaxy clusters, which contain dark matter,

bends and distorts the light of more distant galaxies located behind the cluster. This phenomenon is called gravitational lensing.

Hubble's uniquely sharp vision allows astronomers to map the distribution of dark matter in space using gravitational lensing.

Large galaxy clusters contain both dark matter and normal matter. By observing the areas around massive clusters of galaxies

astronomers can identify warped background galaxies and reverse-engineer their distortions to reveal where the densest

concentrations of matter lie. Mathematical models of these results shed light on the location and properties of the lensing

material, both visible and invisible (dark). The universe appears to have about five times more dark matter than regular matter and seems to be organized around an immense network of dark matter filaments that have grown over time. Massive visible structures, like galaxy clusters, are found at the intersections of these filaments. (Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[Image Captions]

[Image 1 Caption] Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112. Bottom: These two pictures show a massive galaxy cluster, CI 0024+17. The visible-light image (left) shows blue arcs among the yellowish galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue overlay (right) shows the dark matter density needed to account for the gravitational distortions.

Ground Truth Answer

The visible-light image of CI 0024+17 shows blue arc-shaped features among the yellowish cluster galaxies. The text explains these are magnified, distorted images of background galaxies whose light has been bent by the cluster's gravity — a phenomenon called gravitational lensing. The second image overlays in blue the dark matter density distribution required to mathematically account for the observed distortions. The inference chain is: (1) measure the shapes, positions, and degree of distortion of the background arcs; (2) mathematically reverse-engineer the mass distribution needed to produce those specific distortions; (3) compare that total mass to the luminous matter visible in the cluster. The text states the result is that the universe appears to contain about five times more dark matter than regular matter, and that large structures are found at intersections of immense dark matter filament networks. The two images together show both the raw observational input (arcs) and the derived mass product (blue overlay), making the lensing inference chain visually complete.

Model Answer

The provided images and text describe the phenomenon of gravitational lensing, which is used to infer the presence and distribution of dark matter. The text explains that the gravity of massive galaxy clusters, which contain dark matter, bends and distorts the light of more distant galaxies, a phenomenon called gravitational lensing.

Image 1 illustrates this process with the galaxy cluster CI 0024+17. The visible-light image (left) shows blue arcs among the yellowish galaxies, which are described as magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue overlay (right) in this image shows the dark matter density required to account for these gravitational distortions.

The chain of inference connects the visible arc morphology to the dark matter mass estimate as follows: the distortion and bending of the light from background galaxies, visible as blue arcs in the visible-light image, is caused by the gravity of the massive galaxy cluster, which contains dark matter. By observing these warped background galaxies and reverse-engineering their distortions, astronomers can reveal where the densest concentrations of matter lie, which allows them to determine the location and properties of the lensing material, both visible and invisible (dark). Mathematical models based on these results shed light on the location and properties of the lensing material, including the dark matter.

Inference: 23.0185 s · E2E: 23.6013 s · CPU: 0.2% · RAM: 0.093 GB · GPU: 32.1% · Images: 1 · FCD: 16.97

Retrieved Images



Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112. Bottom: These two pictures show a massive galaxy cluster, Cl 0024+17. The visible-light image (left) shows blue arcs among the yellowish galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue overlay (right) shows the dark matter density needed to account for the gravitational distortions.

Query 7 of 14

Question

The image series of Supernova 1987A from 1994 through 2006 shows a ring of material progressively lighting up over twelve years. Using both the time-sequence images and the text description of the shock wave dynamics, explain what physical process the brightening ring documents, why it is observable only over a multi-year baseline, and what this reveals about the pre-explosion mass loss history of the progenitor star.

Retrieved Context

[1] Sept. 24, 1994

Feb. 6, 1998

Mar. 23, 2001

Supernova 1987A as seen by

Hubble in 1994.

Jan. 5, 2003

Dec. 15, 2004

Dec. 6, 2006

A shock wave of material from the central, exploded star of SN 1987A is slamming into an existing ring of material around the doomed star. As it does, various spots in the ring are heated and begin to glow. The nature of the strangely shaped pink feature in the center of the ring is not well understood.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 15)

[2] A shock wave of material from the central, exploded star of SN 1987A is slamming into an existing ring of material around the doomed star. As it does, various spots in the ring are heated and begin to glow. The nature of the strangely shaped pink feature in the center of the ring is not well understood.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 15)

[3] Documenting the Death Throes of Stars

Hubble has revealed unprecedented details in the appearance of Sun-like stars that have entered the death throes of their lives. Ground-based images suggested that many of these objects, called planetary nebulas (though they are unrelated to planets), have simple spherical shapes. Hubble has shown, however, that their shapes are much more varied and complex. Some look like pinwheels, others like butterflies, and still others like hourglasses. Such images yield insights into the complex dynamics that accompany a star's release of its outer gaseous layers before it collapses to form a white dwarf. Turning its sights to the tattered remains of a more massive star's explosive death, Hubble observations of Supernova 1987A revealed three mysterious rings of material encircling the doomed star. The telescope also spied bright spots on the middle ring's inner region caused by an expanding wave of matter slamming into it from the explosion. Likewise, Hubble's view of M1, the Crab Nebula, showed details never before seen about this mighty blast and the rapidly spinning pulsar that remains at its core.

Hubble has revealed the astounding variety and amazing complexity of planetary nebulas.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 14)

[Image Captions]

[Image 1 Caption] Sept. 24, 1994

Feb. 6, 1998

Mar. 23, 2001 Jan. 5, 2003

Dec. 15, 2004

Dec. 6, 2006 A shock wave of material from the central, exploded star of SN 1987A is slamming into an existing ring of material around the doomed star. As it does, various spots in the ring are heated and begin to glow. The nature of the strangely shaped pink feature in the center of the ring is not well understood.

Ground Truth Answer

The six-panel image sequence shows SN 1987A's middle ring going from a faint structure in 1994 to a brightening necklace of hot spots by 2006. The text explains that an expanding shock wave from the central explosion is slamming into a pre-existing ring of material surrounding the dead star. As the blast wave reaches and compresses ring material, those sections are heated and begin to glow — the sequentially brightening spots represent the shock front progressively encountering denser regions around the ring's circumference. The process requires a multi-year baseline because the shock wave must physically travel from the explosion center to the ring, and the ring's finite circumference means different azimuthal sections are reached at slightly different times. The existence of the ring itself implies the progenitor star shed a structured shell of material in an asymmetric mass-loss event before it exploded, providing direct evidence that massive stars undergo significant pre-supernova mass ejection that shapes the circumstellar environment the eventual blast wave must propagate through.

Model Answer

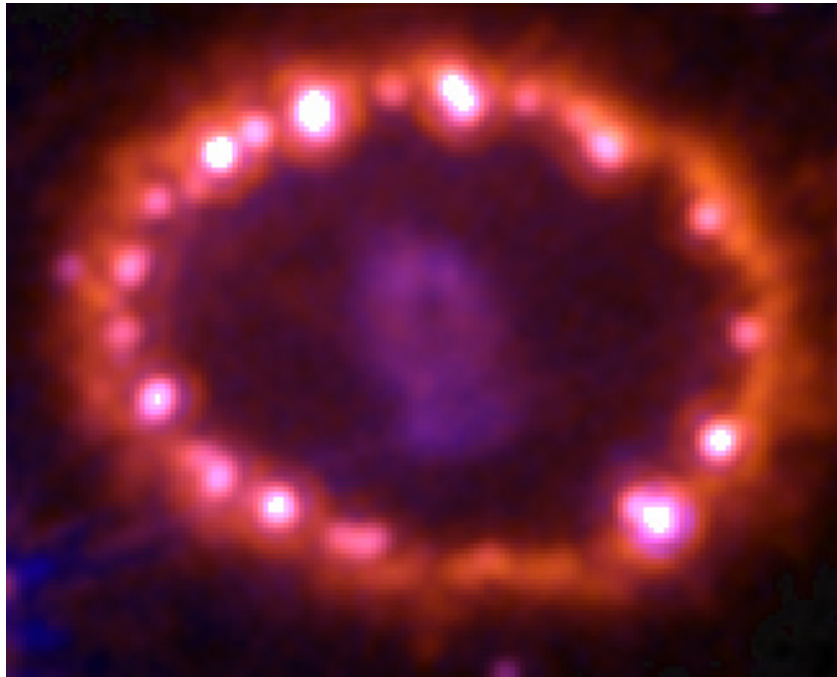
The image series of Supernova 1987A from 1994 through 2006 documents a shock wave of material from the central, exploded star slamming into an existing ring of material around the doomed star. As this interaction occurs, various spots in the ring are heated and begin to glow. This process is described as a shock wave of material from the central, exploded star impacting an existing ring of material.

This brightening ring documents the heating of the ring material caused by the impact of the shock wave. It is observable over a multi-year baseline because the interaction between the shock wave and the ring takes time to manifest as observable changes in the ring's luminosity.

This observation reveals information about the pre-explosion mass loss history of the progenitor star, as the interaction involves the material that existed around the doomed star prior to the supernova event.

Inference: 14.2546 s · E2E: 14.7613 s · CPU: 0.3% · RAM: 0.095 GB · GPU: 30.2% · Images: 1 · FCD: 18.58

Retrieved Images



Sept. 24, 1994 Feb. 6, 1998 Mar. 23, 2001 Jan. 5, 2003 Dec. 15, 2004 Dec. 6, 2006 A shock wave of material from the central, exploded star of SN 1987A is slamming into an existing ring of material around the doomed star. As it does, various spots in the ring are heated and begin to glow. The nature of the strangely shaped pink feature in the center of the ring is not well understood.

Query 8 of 14

Question

The Saturn montage image shows the planet alongside several of its moons at different scales. Using the image to identify visible moons and the text description of Voyager 1's Titan encounter, explain the specific trade-off that made Titan the highest-priority Saturn target, why that priority decision permanently closed Voyager 1's path to the outer planets, and what Voyager 2's different Saturn geometry preserved for the mission as a whole.

Retrieved Context

[1] Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right). Images are not to scale. Courtesy of NASA.

Two more stops: Uranus and Neptune

Voyager 2's flyby of Saturn set it up to automatically fly by Uranus after a coast of five and a half years. It began to study the Uranian system in November 1985 and flew within 50,600 miles of its cloud tops on January 24, 1986, before concluding its observations in February. In conducting the first ever robotic reconnaissance of Uranus, Voyager 2 observed the planet's atmosphere, satellites and thin ring system, discovering 11 new moons and a previously unknown magnetic field and returning about 8,000 pictures. And then, it was on to Neptune, arriving there after another coast of three and a half years. Beginning its study in June 1989, Voyager 2 flew within just 3,076 miles to Neptune's cloud tops on August 25, concluding its observations in October. That close pass to Neptune allowed detailed observations of its largest moon Triton. During this first ever close up study of Neptune, Voyager 2 returned about 10,000 photographs of the planet, its satellites and dark rings, discovering 6 new moons, a Great Dark Spot on (Source: Voyager Grand Tour PDF.pdf, page 3)

[2] Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton (right). Courtesy of NASA.

On February 14, 1990, more than 12 years after it began its journey and shortly before its cameras were permanently turned off to conserve power, Voyager 1 spun around and pointed them back into the solar system. In a mosaic of 60 images, it was able to capture a "family portrait" of six of the solar system's planets, including a pale blue dot called Earth. It was fitting that these were the last pictures from either Voyager.

(Source: Voyager Grand Tour PDF.pdf, page 4)

[3] Both Voyagers quietly coasted as they headed toward their next target, the ringed planet Saturn. Once again, Voyager 1 was first, beginning observations in August 1980 before making its closest approach 77,000 miles above Saturn's cloud tops on November 12, and finishing studies in December.

Significantly, Voyager 1 made a close flyby of Saturn's large moon Titan, larger than our own Moon and with a dense atmosphere, coming within 4,000 miles of its surface near its South Pole. The Titan encounter changed Voyager 1's trajectory so it could not make any further planetary encounters – thus beginning its long journey out of our solar system. Voyager 2 started its examination of Saturn in June 1981, making its closest approach at 63,000 miles on August 25, and finishing observations in September.

The instruments aboard the Voyagers returned a wealth of new information about Saturn and the cameras returned thousands of images of the planet, its rings and satellites. Among significant discoveries, the Voyagers found that Saturn's rings were far more complex than previously believed, Titan may have lakes of liquid hydrocarbons, Saturn has auroras near its poles much like Earth does, and that there were three previously unknown satellites.

(Source: Voyager Grand Tour PDF.pdf, page 3)

[Image Captions]

[Image 1 Caption] Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right). Images are not to scale. Courtesy of NASA.

Ground Truth Answer

The Saturn montage shows the planet's rings and several moons, with Titan visible as a large body. The text states Titan is larger than Earth's own Moon and was known to have a dense atmosphere, making it uniquely scientifically valuable among Saturn's satellites. Voyager 1 flew within 4,000 miles of Titan's surface near its South Pole — the text explicitly states this Titan encounter changed Voyager 1's trajectory so it could not make any further planetary encounters. Voyager 2's closest Saturn approach was at 63,000 miles, a different geometry that did not incur the same trajectory penalty. The consequence was that Voyager 2's Saturn flyby automatically set up a coast of five and a half years to Uranus, where it discovered 11 new moons, and then continued to Neptune where it discovered 6 more moons and observed geysers on Triton. The two-spacecraft strategy deliberately split the objectives: one vehicle maximized Titan science at the cost of the extended tour, while the other preserved the grand-tour continuation.

Model Answer

The Saturn montage image displays the planet alongside several of its moons at different scales. The text describes that Voyager 1 made a close flyby of Saturn's large moon Titan, coming within 4,000 miles of its surface near its South Pole, and this encounter changed Voyager 1's trajectory so it could not make any further planetary encounters, thus beginning its long journey out of our solar system. This encounter made Titan a significant event, as it was a large moon with a dense atmosphere, and the close pass allowed detailed observations of Titan.

The text does not explicitly state why Titan was the highest-priority Saturn target, nor does it detail what Voyager 2's different Saturn geometry preserved for the mission as a whole.

Inference: 14.1044 s · E2E: 14.5410 s · CPU: 0.3% · RAM: 0.097 GB · GPU: 25.1% · Images: 1 · FCD: 25.18

Retrieved Images



Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right). Images are not to scale. Courtesy of NASA.

Query 9 of 14

Question

The Uranus montage and the Neptune-Triton montage each show a planet alongside several moons. Using these images in combination with the text on Voyager 2's encounters, explain why Voyager 2 discovered 11 moons at Uranus and 6 at Neptune despite having far less observing time at each world than it had spent at Jupiter, and what this implies about the relationship between mission geometry and small-body discovery yield.

Retrieved Context

[1] Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton (right). Courtesy of NASA.

On February 14, 1990, more than 12 years after it began its journey and shortly before its cameras were permanently turned off to conserve power, Voyager 1 spun around and pointed them back into the solar system. In a mosaic of 60 images, it was able to capture a “family portrait” of six of the solar system’s planets, including a pale blue dot called Earth. It was fitting that these were the last pictures from either Voyager.

(Source: Voyager Grand Tour PDF.pdf, page 4)

[2] Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right). Images are not to scale. Courtesy of NASA.

Two more stops: Uranus and Neptune

Voyager 2’s flyby of Saturn set it up to automatically fly by Uranus after a coast of five and a half years. It began to study the Uranian system in November 1985 and flew within 50,600 miles of its cloud tops on January 24, 1986, before concluding its observations in February. In conducting the first ever robotic reconnaissance of Uranus, Voyager 2 observed the planet’s atmosphere, satellites and thin ring system, discovering 11 new moons and a previously unknown magnetic field and returning about 8,000 pictures. And then, it was on to Neptune, arriving there after another coast of three and a half years. Beginning its study in June 1989, Voyager 2 flew within just 3,076 miles to Neptune’s cloud tops on August 25, concluding its observations in October. That close pass to Neptune allowed detailed observations of its largest moon Triton. During this first ever close up study of Neptune, Voyager 2 returned about 10,000 photographs of the planet, its satellites and dark rings, discovering 6 new moons, a Great Dark Spot on

[3] the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the Neptune system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid Voyagers weren’t quite done yet.

(Source: Voyager Grand Tour PDF.pdf, page 4)

[Image Captions]

[Image 1 Caption] the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the Neptune

system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid Voyagers weren’t quite done yet. Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton (right). Courtesy of NASA.

Ground Truth Answer

The Uranus montage shows several larger moons around the tilted ice giant, while the Neptune montage shows Triton prominently alongside Neptune. The text states Voyager 2 began studying Uranus in November 1985 and made its closest approach on January 24, 1986, discovering 11 new moons, while the Neptune encounter beginning June 1989 yielded 6 new moons. At Jupiter, which Voyager 2 studied intensively from April to August 1979 — a much longer campaign — only three previously undiscovered small moons were found. The inversion in discovery

yield relative to time suggests that proximity geometry and illumination conditions during a fast flyby can be more favorable for detecting small, faint satellites than a longer but more distant observation campaign. At Uranus and Neptune the extremely close approach distances (50,600 miles and 3,076 miles respectively) provided the angular resolution and lighting angles needed to detect objects that would have remained hidden from farther distances, regardless of integration time.

Model Answer

The provided text and images detail the Voyager 2 flyby of Uranus and Neptune, which involved close observations of the planets, their atmospheres, satellites, and ring systems. Voyager 2 discovered 11 new moons around Uranus and 6 new moons around Neptune during these encounters. This discovery occurred despite having less observing time at Uranus and Neptune compared to the time spent at Jupiter.

The text indicates that Voyager 2's close pass to Neptune allowed for detailed observations of its largest moon Triton, and during this study, it discovered 6 new moons and a Great Dark Spot. Similarly, the observation of Uranus included the discovery of 11 new moons. This suggests that the geometry of the encounter, specifically the close flybys, was instrumental in these discoveries. The text notes that Voyager 2 flew within a certain distance of Uranus's cloud tops and Neptune's cloud tops, which facilitated these observations. This implies that the specific trajectory and proximity during the flybys, rather than just the total observing time, significantly influenced the yield of small-body discoveries.

Inference: 17.8926 s · E2E: 18.4082 s · CPU: 0.3% · RAM: 0.099 GB · GPU: 29.4% · Images: 1 · FCD: 18.50

Retrieved Images



the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the Neptune system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid Voyagers weren't quite done yet. Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton (right). Courtesy of NASA.

Query 10 of 14

Question

Curiosity's self-portrait assembled from MAHLI images is shown in the document. Using the image to assess what the composite reveals about rover configuration and condition, and the technical text describing MAHLI's capabilities and arm placement, explain why this imaging method provides more diagnostic value for mission engineers than a single wide-angle shot from a fixed camera would.

Retrieved Context

[1] The mission uses radio relays via Mars orbiters as the principal means of communication between Curiosity and the Deep Space Network's antennas on Earth. In the first two years after Curiosity's landing, the orbiters downlinked 48 gigabytes of data from Curiosity.

In April 2004, NASA solicited proposals for specific instruments and investigations to be carried by Mars Science Laboratory. The agency selected eight of the proposals later that year and also reached agreements with Russia and Spain to carry instruments those nations provided. A suite of instruments named Sample Analysis at Mars analyzes samples of material collected and delivered by A self-portrait of Curiosity, built up from pictures taken by its Mars Hand Lens Imager.

the rover's arm, plus atmospheric samples. It includes a gas chromatograph, a mass spectrometer and a tunable drive, a rocker-bogie suspension system, and cameras laser spectrometer with combined capabilities to identify a mounted on a mast to help the mission's team on Earth wide range of carbon-containing compounds and determine the ratios of different isotopes of key elements. Iso-rovers, Curiosity carries equipment to gather and process tope ratios are clues to understanding the history of Mars' samples of rocks and soil, distributing them to onboard atmosphere and water. The principal investigator is Paul test chambers inside analytical instruments.

Mahaffy of NASA's Goddard Space Flight Center, Green-NASA's Jet Propulsion Laboratory (JPL), Pasadena, Calif., builder of the Mars Science Laboratory, engineered Curiosity to roll over obstacles up to 65 centimeters (25 inches) high and to travel up to about 200 meters (660 feet) (Source: mars-science-laboratory.pdf, page 3)

[2] samples of rocks and soil, distributing them to onboard atmosphere and water. The principal investigator is Paul test chambers inside analytical instruments.

Mahaffy of NASA's Goddard Space Flight Center, Green-NASA's Jet Propulsion Laboratory (JPL), Pasadena, Calif., builder of the Mars Science Laboratory, engineered Curiosity to roll over obstacles up to 65 centimeters (25 inches) high and to travel up to about 200 meters (660 feet) (Source: mars-science-laboratory.pdf, page 3)

osity to roll over obstacles up to 65 centimeters (25 inches) high and to travel up to about 200 meters (660 feet) per day on Martian terrain.

rocks and soils, and to measure bulk composition. The principal investigator is David Blake of NASA's Ames Research Center, Moffett Field, Calif.

The rover's electrical power is supplied by a U.S. Department of Energy radioisotope power generator. The multi-mission radioisotope thermoelectric generator produces electricity from the heat of plutonium-238's radioactive decay. Mounted on the arm, the Mars Hand Lens Imager takes extreme close-up pictures of rocks and soil, revealing details smaller than the width of a human hair. It can also operate for a full Mars year (687 Earth days) or more. At launch, the generator provided about 110 watts of electrical power to operate the portraits of Curiosity. The principal investigator is Kenneth Edgett of Malin Space Science Systems, San Diego. Warm fluids heated by the generator's excess heat (Source: mars-science-laboratory.pdf, page 3)

[3] details smaller than the width of a human hair. It can also operate for a full Mars year (687 Earth days) or more. At launch, the generator provided about 110 watts of electrical power to operate the portraits of Curiosity. The p

[Image Captions]

[Image 1 Caption] A self-portrait of Curiosity, built up from pictures taken by its Mars Hand Lens Imager. drive, a rocker-bogie suspension system, and cameras mounted on a mast to help the mission's team on Earth select exploration targets and driving routes. Unlike earlier

Ground Truth Answer

The self-portrait shows the full rover body, mast, arm, and wheel configuration in a composite assembled from many close-range images. The text states MAHLI is the Mars Hand Lens Imager mounted on the robotic arm, capable of taking extreme close-up images revealing details smaller than a human hair, and that it can focus on hard-to-reach objects more than an arm's length away. Because the arm repositions the camera at multiple locations and angles, the resulting mosaic provides overlapping, close-range coverage of surfaces that a fixed wide-angle camera would capture only at low resolution and from a single viewpoint. For engineers, this means dust accumulation on solar panel analogs, wheel wear patterns, hardware surface changes, and instrument condition can all be assessed at sub-millimeter detail in routine imaging without requiring dedicated inspection hardware. The method also confirms arm articulation health implicitly — a successful multi-position mosaic is itself evidence of normal arm function.

Model Answer

The self-portrait of Curiosity, built from pictures taken by its Mars Hand Lens Imager, reveals the rover's configuration, including its arm, drive, rocker-bogie suspension system, and cameras mounted on a mast. The technical text describes the Mars Hand Lens Imager as providing "close-up pictures of rocks and soil, revealing details smaller than the width of a human hair." Furthermore, the text notes that the instrument is mounted on the arm and can focus on hard-to-reach objects more than an arm's length. This capability, combined with the assembled self-portrait, provides more diagnostic value for mission engineers than a single wide-angle shot from a fixed camera because it offers detailed, close-up information about the rover's physical state and the details of the terrain. The composite image allows engineers to assess the rover's configuration and condition in a way that a single, fixed-angle shot cannot, by providing visual evidence of the rover's physical structure and the ability of the imaging system to focus on fine details.

Inference: 18.0734 s · E2E: 18.6378 s · CPU: 0.3% · RAM: 0.106 GB · GPU: 27.9% · Images: 1 · FCD: 30.38

Retrieved Images



A self-portrait of Curiosity, built up from pictures taken by its Mars Hand Lens Imager. drive, a rocker-bogie suspension system, and cameras mounted on a mast to help the mission's team on Earth select exploration targets and driving routes. Unlike earlier

Query 11 of 14

Question

The Cepheid star brightness sequence images from the Andromeda galaxy and the distant supernova comparison images from 1995 and 2002 are presented as two complementary distance-measurement methods. Using both image sets and the text about refining the Hubble constant, explain why astronomers need both techniques rather than relying on only one, and how the combination of refined distances and velocity measurements changed the estimated age of the universe.

Retrieved Context

[1] Above: Astronomers use cyclical changes in the brightness of Cepheid stars to determine astronomical distances. The arrow points to a Cepheid star in the Andromeda galaxy observed by Hubble (inset boxes). Right: Certain supernovas have a characteristic maximum brightness that can be used to calculate their distances from Earth. Refining celestial distances enables astronomers to better calculate the expansion rate of the universe. The arrow points to a distant supernova discovered in an area of the sky first imaged in 1995 called the Hubble Deep Field. Astronomers found the supernova when they targeted the same area of sky again in 2002 and saw a change.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 3)

[2] Tracing the Growth of Galaxies
Like documenting a child's development in a scrapbook, astronomers have used Hubble to capture the appearance of many developing galaxies throughout cosmic time. This is possible because of the mathematical relationship between cosmic distance and time: the deeper Hubble peers into space, the farther back it looks in time. As it happens, the most distant and earliest galaxies spied by Hubble are smaller and more irregularly shaped than today's grand spiral and elliptical galaxies. This is evidence that galaxies grew over time through mergers with other galaxies to become the giant systems we see today.
amounts of stars and gas in galaxies, the types and amounts of identifiable chemical elements present, and star-formation rates. And the evolution continues. Hubble observations of our neighboring Andromeda galaxy (M31) have allowed astronomers to predict with certainty that a titanic collision between our Milky Way galaxy and Andromeda will inevitably take place beginning 4 billion years from now. The galaxy is now 2.5 million light-years away, but it is inescapably moving toward the Milky Way under the mutual pull of gravity between the two galaxies and the invisible dark matter that surrounds them both. The merger will likely result in the creation of a giant elliptical galaxy billions of years from now.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 4)

[3] Discovering a Runaway Universe

Our cosmos is getting bigger. Nearly a century ago Edwin Hubble measured the expansion rate of the universe. This value, called the Hubble constant, is an essential ingredient needed to determine the age, size, and fate of the cosmos. Before Hubble was launched, the value of the Hubble constant was imprecise, and calculations of the universe's age ranged from 10 billion to 20 billion years. Now, astronomers using Hubble have refined their estimates of the universe's present expansion rate and are working to make it more accurate. They do this by getting better galaxy distance measurements from Hubble and coupling these values with the best galaxy-velocity measurements obtained from other telescopes. Scientists measure distances by comparing the brightness of a known object in our galaxy (like a star or an exploded star) to that of a similar object in a distant galaxy. With Hubble's refined distance values, calculations currently put the age of the universe as 13.8 billion years.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 3)

[Image Captions]

[Image 1 Caption] Background photo: R. Gendler Above: Astronomers use cyclical changes in the brightness of Cepheid stars to determine astronomical distances. The arrow points to a Cepheid star in the Andromeda galaxy observed by Hubble (inset boxes).

Ground Truth Answer

The Cepheid sequence shows a star in Andromeda varying in brightness across multiple observation dates — this cyclical brightness change is the basis for the period-luminosity distance calibration. The 1995 and 2002 comparison images show a supernova appearing in a previously empty patch of the Hubble Deep Field, illustrating how Type Ia supernovas serve as standard candles at distances where individual Cepheids cannot be resolved. The text explains astronomers measure distances by comparing the known brightness of objects like Cepheid stars or standard supernovas to their apparent brightness in distant galaxies. Cepheids are well-resolved and reliable within the local universe but too faint to detect in very distant galaxies; supernovas are bright enough to see at cosmological distances but need Cepheid-based calibration in the local universe to set their absolute luminosity scale. By combining both ladders, Hubble refined galaxy distance measurements sufficiently to narrow the Hubble constant uncertainty, and coupling those distances with galaxy velocity measurements from other telescopes, calculations currently place the age of the universe at 13.8 billion years, far more precise than the pre-Hubble range of 10 to 20 billion years.

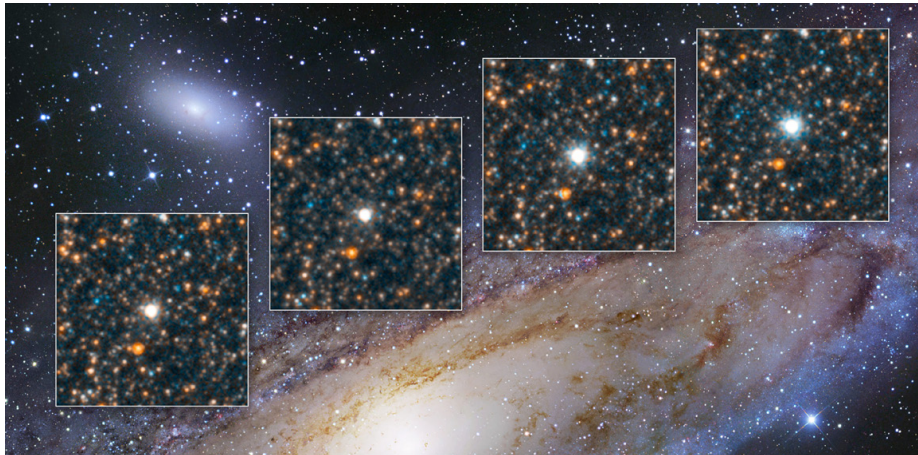
Model Answer

Astronomers utilize both the Cepheid star brightness sequence images and the distant supernova comparison images as complementary distance-measurement methods to refine astronomical understanding. The text indicates that astronomers use cyclical changes in the brightness of Cepheid stars to determine astronomical distances, and certain supernovas have a characteristic maximum brightness used to calculate their distances from Earth.

Refining celestial distances enables astronomers to better calculate the expansion rate of the universe. Furthermore, the text states that astronomers refine estimates of the universe's present expansion rate by getting better galaxy distance measurements from Hubble and coupling these values with the best galaxy-velocity measurements obtained from other telescopes. This process of using refined distance values and velocity measurements has led to a revised calculation, currently placing the age of the universe as 13.8 billion years, which was an improvement over previous estimates ranging from 10 billion to 20 billion years.

Inference: 15.1981 s · E2E: 15.7606 s · CPU: 0.3% · RAM: 0.114 GB · GPU: 28.6% · Images: 1 · FCD: 16.03

Retrieved Images



Background photo: R. Gendler Above: Astronomers use cyclical changes in the brightness of Cepheid stars to determine astronomical distances. The arrow points to a Cepheid star in the Andromeda galaxy observed by Hubble (inset boxes).

Query 12 of 14

Question

The M84 galaxy images show a camera view with a dark dust band and a spectrograph plot with dramatic color shifts from blue to red. Using both images and the text on black hole mass measurements, explain exactly what physical phenomenon the color shift encodes, why the spectrograph slit position relative to the core is critical to the measurement, and how this technique established the mass relationship between black holes and their host galaxies.

Retrieved Context

[1] Shown here are two Hubble views of the galaxy M84. Left: This camera view shows the bright core at the center of the galaxy surrounded by a (vertical) dark band of gas and dust. Right: This plot was generated by passing light near the core of the galaxy through a spectrograph. The thin, vertical rectangle in the center of the left panel shows the size and shape of the spectrograph's sampling slit. Spectra taken left and right of the slit's position, which is centered on the core of the galaxy, show a dramatic shift in color to blue or red. Blueshifted light indicates that the emitting light source is moving toward Earth, while redshifted light indicates objects that are moving away. At 880,000 miles per hour, stars and glowing gases nearest to the core of M84 are moving the fastest. They are circling a black hole at the center of the galaxy, and so appear to be moving rapidly toward Earth on the left and receding on the right.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 7)

[2] Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112. Bottom: These two pictures show a massive galaxy cluster, Cl 0024+17. The visible-light image (left) shows blue arcs among the yellowish galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue overlay (right) shows the dark matter density needed to account for the gravitational distortions.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[3] Combining images with data from Hubble's spectrographs, researchers have peered into the center of many galaxies and established the existence of large black holes. In a census performed by Hubble in the late 1990s, galaxies NGC 3379 and NGC 3377 were found to have black holes that "weighed in" at 50 million and over 100 million solar masses, respectively, and NGC 4486B was revealed to have a double nucleus at its core.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 7)

[Image Captions]

[Image 1 Caption] Realizing Monster Black Holes Are Everywhere

Ground Truth Answer

The camera image shows M84's bright core crossed by a dark vertical band of gas and dust, while the spectrograph plot shows a sharp transition from blue on one side of the core to red on the other. The text explains that blueshifted light indicates the emitting source is moving toward Earth while redshifted light indicates recession — stars and gas nearest the core are orbiting a central black hole at 880,000 miles per hour, so material on one side of the rotation

axis approaches Earth while material on the other side recedes. The spectrograph slit must be positioned centered on the core because the Doppler shift gradient is steepest where the orbital velocities are highest, right at the innermost region around the black hole. The width and sharpness of the blue-to-red transition encodes the rotational velocity, which through Kepler's laws yields the enclosed mass. Applying this technique across many galaxies in a Hubble census revealed that black hole mass scales with the mass of the host galaxy's central stellar bulge — the larger the galaxy, the larger the black hole — suggesting black holes and their galaxies co-evolved.

Model Answer

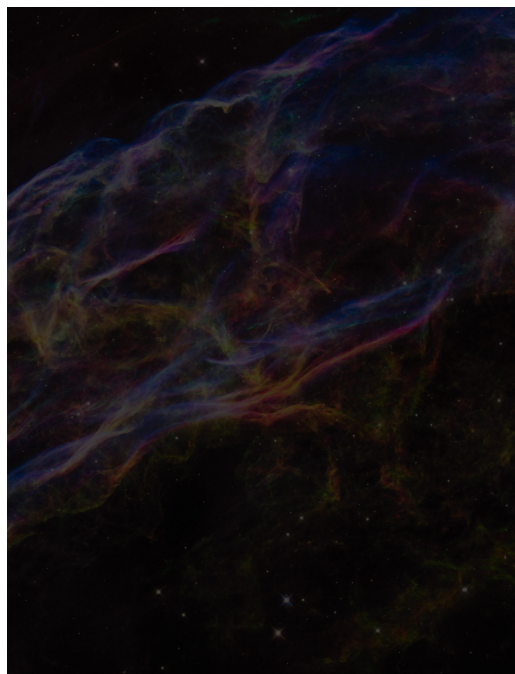
The images and text describe how light from the core of the galaxy M84 is used to infer the motion of the emitting sources, which encodes the physical phenomenon of redshift or blueshift. The text explains that blueshifted light indicates that the emitting light source is moving toward Earth, while redshifted light indicates objects that are moving away.

The spectrograph plot, generated by passing light near the core of the galaxy through a spectrograph, shows a dramatic shift in color to blue or red on either side of the sampling slit, which is centered on the core of the galaxy. This shift is directly related to the motion of the light source. The text further indicates that stars and glowing gases nearest to the core of M84 are moving at 880,000 miles per hour, circling a black hole at the center of the galaxy, and thus appear to be moving rapidly toward Earth on the left (blueshift) and receding on the right (redshift).

The position of the spectrograph slit relative to the core is critical because the spectra taken to the left and right of the slit's position, which is centered on the core of the galaxy, show these shifts. This technique, combined with the knowledge that these motions are caused by objects circling a black hole, established a relationship between the observed color shifts and the presence of black holes, as researchers have peered into the center of many galaxies to establish the existence of large black holes.

Inference: 26.1959 s · E2E: 26.8598 s · CPU: 0.2% · RAM: 0.120 GB · GPU: 31.2% · Images: 1 · FCD: 12.27

Retrieved Images



Realizing Monster Black Holes Are Everywhere

Query 13 of 14

Question

The Jupiter impact sequence showing cloud darkening from Comet Shoemaker-Levy 9, the close-up plume image from the first large fragment impact, and the Saturn aurora image are all on the same or adjacent pages. Using these three images and the text covering both the impact events and aurora observations, explain what each image type contributes uniquely to understanding planetary atmospheric dynamics, and why continuous ultraviolet monitoring capability is the common prerequisite for all three scientific results.

Retrieved Context

[1] This image series, starting in the lower right corner, shows the darkening of Jupiter's clouds as a result of the impact of Comet Shoemaker-Levy 9.

Hubble captured a colossal plume rising above the limb of Jupiter from the impact of the first large fragment of Comet Shoemaker-Levy 9.

(Times along the left side are in hour:minute format.)

Jupiter's moons also have yielded important clues in the search for life beyond Earth. Hubble provided the best evidence yet for an underground saltwater ocean on Ganymede, the largest moon in the solar system, by detecting related activity in Ganymede's own auroras. This subterranean ocean is thought to have more water than all the water on Earth's surface.

Hubble also recorded evidence of transient changes in the atmosphere above the surface of Jupiter's moon Europa.

Astronomers suspect that these disturbances are caused by gas plumes expelled from a subsurface ocean. Identifying liquid water is crucial in the search for habitable worlds beyond Earth and in the quest to find life as we know it.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 8)

[2] Studying the Outer Planets and Moons

Hubble has witnessed impacts on Jupiter that were produced by minor bodies in the solar system. The latest collision observed by Hubble occurred in 2009 when a suspected asteroid plunged into Jupiter's atmosphere, leaving a temporary dark feature the size of the Pacific Ocean. In 1994 Hubble watched 21 fragments of Comet Shoemaker-Levy 9 bombard the giant planet sequentially—the first time astronomers witnessed such an event. Each impact left a temporary black, sooty scar within Jupiter's clouds.

Jupiter is well known for its Great Red Spot, a giant storm roughly the size of Earth that has been continuously visible since at least the late 1800s. The mammoth storm has been shrinking in size for at least 80 years. Astronomers now use Hubble regularly to measure the Red Spot's size and investigate why it is slowly disappearing.

Hubble also made the first images of bright auroras at the northern and southern poles of Saturn and Jupiter.

Auroras are brilliant curtains of light that appear in the upper atmosphere of a planet with a magnetic field. They develop when electrically charged particles trapped in the magnetic field spiral inward at high speeds toward the north and

south magnetic poles. When these particles hit the upper atmosphere, they excite atoms and molecules there causing them to glow in a similar process to that of a fluorescent light. This image series, starting in the lower right corner, shows the darkening of Jupiter's clouds as a result of the impact of Comet Shoemaker-Levy 9.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 8)

[3] Top left: Jupiter's trademark Great Red Spot—a swirling, 300-mile-per-hour storm—has shrunk to its smallest size ever. Astronomers have followed this gradual downsizing since the 1930s and are now keenly monitoring its shrinkage and investigating its cause with Hubble. Top right: Hubble's ability to capture ultraviolet light revealed a region of glowing auroras over the pole of Saturn. While sharing some features of Earth's auroras and others of Jupiter's, Saturn's auroras are different from both and are likely unique in the solar system. Below left: The blue areas mapped on Jupiter's moon Europa mark spots where Hubble found spectroscopic evidence of oxygen and hydrogen—the two elements that form the water molecule. Scientists believe this is most likely the result of water vapor plumes erupting from beneath the surface, with recent visual evidence of these plumes also seen by Hubble (below right). The pictures of Europa are a combination of shots gathered by NASA's Voyager and Galileo missions.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 9)

[Image Captions]

[Image 1 Caption] Studying the Outer Planets and Moons This image series, starting in the lower right corner, shows the darkening of Jupiter's clouds as a result of the impact of Comet Shoemaker-Levy 9. Hubble captured a colossal plume rising above the limb of Jupiter from the impact of the first large fragment of Comet Shoemaker-Levy 9. (Times along the left side are in hour:minute format.)

Ground Truth Answer

The Shoemaker-Levy 9 cloud darkening sequence shows Jupiter's atmosphere progressively accumulating impact scars as 21 fragments struck sequentially in 1994 — each dark sooty scar documents energy deposition depth and lateral spread in the Jovian cloud deck. The plume image from the first large fragment shows a colossal debris column rising above Jupiter's limb in time-stamped frames at minute-level cadence, capturing the transient ejecta dynamics during the first seconds to minutes after impact. The Saturn aurora image shows UV-bright curtains of light at Saturn's polar regions, capturing magnetospheric particle precipitation into the upper atmosphere. Each contributes something distinct: the scar sequence measures integrated impact effects over days to weeks; the plume sequence captures real-time ballistic ejecta dynamics; the aurora image characterizes steady-state magnetospheric coupling. All three depend on ultraviolet sensitivity because: impact plumes and atmospheric ejecta are bright in UV; auroral emission from excited atmospheric molecules peaks in the UV; and the temporal fidelity needed to capture transients requires space-based UV capability free from atmospheric UV absorption that blocks ground-based observation.

Model Answer

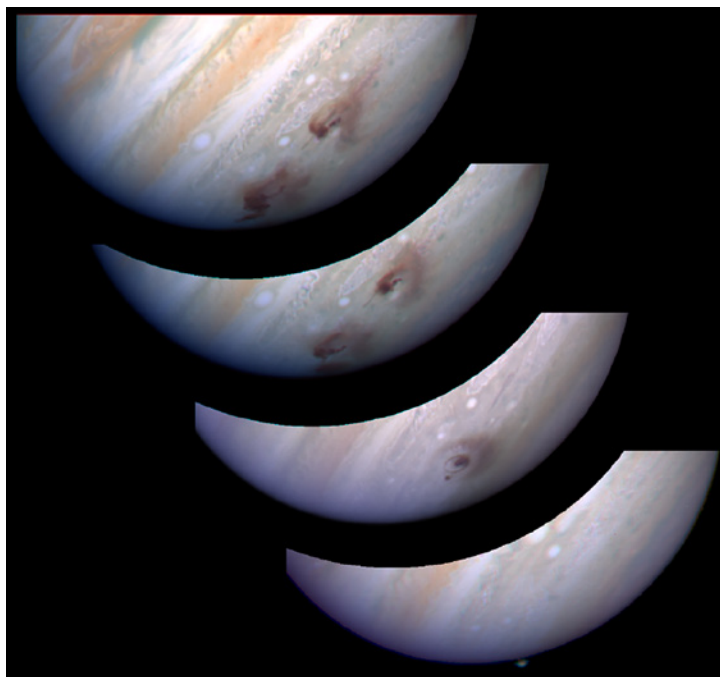
The image series showing the darkening of Jupiter's clouds due to the impact of Comet Shoemaker-Levy 9, the close-up plume image from the first large fragment impact, and the Saturn aurora image each contribute uniquely to understanding planetary atmospheric dynamics. The image series of Jupiter's cloud darkening demonstrates the direct physical effect of the impact on the atmosphere. The close-up plume image shows a colossal plume rising above the limb of Jupiter resulting from the impact of the first large fragment of Comet Shoemaker-Levy 9. The

Saturn aurora image reveals the presence of bright auroras at the poles of Saturn, which are caused by electrically charged particles interacting with the upper atmosphere due to the planet's magnetic field.

Continuous ultraviolet monitoring capability is the common prerequisite for all three scientific results because the text indicates that Hubble captured these events, and the ability to capture ultraviolet light is essential for revealing these phenomena. For instance, the text notes that Hubble's ability to capture ultraviolet light revealed the auroras over Saturn's pole. Furthermore, the text discusses how astronomers use Hubble to measure the Great Red Spot's size and investigate its disappearance, and the context implies that this capability is central to observing the atmospheric changes and related phenomena, such as the gas plumes suspected to be caused by subsurface oceans on Europa.

Inference: 23.4039 s · E2E: 23.9639 s · CPU: 0.2% · RAM: 0.116 GB · GPU: 32.2% · Images: 1 · FCD: 17.43

Retrieved Images



Studying the Outer Planets and Moons This image series, starting in the lower right corner, shows the darkening of Jupiter's clouds as a result of the impact of Comet Shoemaker-Levy 9. Hubble captured a colossal plume rising above the limb of Jupiter from the impact of the first large fragment of Comet Shoemaker-Levy 9. (Times along the left side are in hour:minute format.)

Query 14 of 14

Question

The TW Hydrae disk image shows a gap structure around the star, and the 1997 and 2012 Beta Pictoris disk images show an edge-on disk changing over fifteen years. Using both sets of images alongside the planet-formation text, explain what specific morphological features in each image constitute evidence for an embedded or nearby planet, why temporal baseline matters for the Beta Pictoris case but not for TW Hydrae, and how these two systems together support a dynamic rather than static picture of disk evolution.

Retrieved Context

[1] This Hubble image and illustration show a gap in a protoplanetary disk of dust and gas around the nearby star TW Hydrae. The gap's presence is probably due to the effects of a growing, unseen planet that is gravitationally sweeping up material and carving out a lane in the disk. Astronomers used a masking device within the Hubble camera to block out the star's bright light so that the disk's structure could be seen.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 17)

[2] Finding Planetary Construction Zones
Astronomers used Hubble to confirm that planets form in dust disks around stars. The telescope first resolved protoplanetary disks around nearly 200 stars in the bright Orion Nebula. Looking at nearby stars elsewhere in the sky, Hubble completed the largest and most sensitive visible-light imaging survey of dusty debris disks, which were probably created by collisions between leftover objects from planet formation. Two particular stars illustrate these findings: TW Hydrae and Beta Pictoris. Using a mask to block the star's bright light, Hubble scientists spotted a mysterious gap in a vast protoplanetary disk of gas and dust swirling around the star TW Hydrae. The gap is most likely caused by a growing, unseen planet that is gravitationally sweeping up material and carving out a lane in the disk like a snowplow. It is 1.9 billion miles wide and not yet completely cleared of material. Researchers have also noted changes in the planetary disk surrounding Beta Pictoris. By masking out the star's light, scientists have studied changes in the orbiting material caused by a massive planet embedded within its dust disk. Astronomers spotted the planet using the European Southern Observatory. Here are a few of the many protoplanetary disks that were discovered by Hubble in the large, majestic Orion Nebula. The disks appear as silhouettes, illuminated from behind by the glowing gases of the nebula. Below left and right: Hubble has provided detailed pictures showing changes in the large, edge-on, gas-and-dust disk encircling the 20-million-year-old star Beta Pictoris.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 17)

[3] Below left and right: Hubble has provided detailed pictures showing

changes in the large, edge-on, gas-and-dust disk encircling the 20-million-year-old star Beta Pictoris.

(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 17)

[Image Captions]

[Image 1 Caption] Below left and right: Hubble has provided detailed pictures showing changes in the large, edge-on, gas-and-dust disk encircling the 20-million-year-old star Beta Pictoris.

Ground Truth Answer

The TW Hydrae image shows a protoplanetary disk with a gap approximately 1.9 billion miles wide that is not yet completely cleared of material. The text explains this gap is most likely caused by a growing, unseen planet gravitationally sweeping up material like a snowplow, carving a lane in the disk. The morphological evidence is the gap itself — its width, incomplete clearing, and position within the disk are all consistent with an embedded planet at that orbital radius. The Beta Pictoris pair shows the large edge-on disk in 1997 and 2012; the text states scientists studied changes in the orbiting material caused by a massive planet embedded within the dust disk. For TW Hydrae, a single image suffices because the gap is a static structural feature that persists on timescales far longer than an observation campaign. For Beta Pictoris, the 15-year baseline is essential because the planet-induced changes in disk structure — warps, asymmetries, or material redistribution — are only detectable as differences between epochs, not from any single snapshot. Together, TW Hydrae shows a planet actively carving its initial gap while Beta Pictoris shows a more evolved system where a mature embedded planet continues to dynamically reshape disk material over decadal timescales.

Model Answer

The image of the TW Hydrae disk shows a gap in the protoplanetary disk of dust and gas around the star, which is probably due to the effects of a growing, unseen planet gravitationally sweeping up material and carving out a lane in the disk. This gap is evidence for an embedded planet.

In contrast, the images of the Beta Pictoris disk show changes in the large, edge-on, gas-and-dust disk encircling the star over time, illustrating changes in the orbiting material caused by a massive planet embedded within its dust disk.

The temporal baseline matters for the Beta Pictoris case because scientists have studied changes in the orbiting material over time, which is indicative of dynamic evolution. For the TW Hydrae case, the text describes the gap as a feature caused by a growing planet, suggesting a dynamic process. Together, these two systems support a dynamic rather than static picture of disk evolution by showing different manifestations of planetary influence: a carved-out gap in TW Hydrae and observable changes in the disk structure around Beta Pictoris, both pointing to the dynamic interaction between a star and orbiting planets within their protoplanetary disks.

Inference: 18.9619 s · E2E: 19.3820 s · CPU: 0.2% · RAM: 0.117 GB · GPU: 30.1% · Images: 1 · FCD: 12.84

Retrieved Images



Below left and right: Hubble has provided detailed pictures showing changes in the large, edge-on, gas-and-dust disk encircling the 20-million-year-old star Beta Pictoris.